

Cactus Product Growth Research

Customer evidence, Cactus 1.0 and 2.0 product analytics, growth strategy, and an outcome-based roadmap

Planning edition: July 27, 2026

Public redacted edition - direct customer identities, private recordings, account-level data, and authenticated analytics links are excluded.

Cactus Growth and Revenue Product Plan

Planning date: 2026-07-27

Planning horizon: 12 months, with conditional bets beyond 12 months

Source evidence: 946 external Sybill meetings, Cactus 1.0 and 2.0 PostHog, the existing 10-project opportunity portfolio, and the 60-ticket planning backlog

Product-version evidence convention

Every PostHog metric in this plan must be read through one of three explicit lenses:

- Cactus 1.0 - legacy behavior benchmark: evidence of jobs users repeatedly performed before the refactor. It validates demand and habitual behavior, but it does not describe the current Cactus 2.0 experience.
- Cactus 2.0 - current-product evidence: evidence of the refactored product's activation, workflow adoption, reliability, collaboration, and scalability. It is the primary baseline for current UX and growth decisions.
- Cross-version - migration evidence: a comparison used to test whether an important Cactus 1.0 job has been successfully carried into Cactus 2.0. Event taxonomies and route definitions are not assumed to be identical, so these comparisons are directional unless equivalence has been verified.

Both products are intentionally analyzed. Cactus 1.0 identifies proven jobs and usage intensity; Cactus 2.0 shows whether the more scalable redesign makes those jobs easier to discover, complete, trust, and repeat.

Product	PostHog project	Role in decisions
Cactus 1.0	Project 98643	Legacy demand and behavior benchmark
Cactus 2.0	Project 237852	Current experience, activation, reliability, and adoption baseline

The decline of activity in Cactus 1.0 during migration must not be interpreted by itself as declining customer demand or churn. Nor should active-user counts from the two projects be added without cross-project identity resolution.

The refactor should be evaluated against four questions:

1. Preservation: did Cactus 2.0 carry forward the high-value jobs proven in Cactus 1.0?
2. Improvement: are those jobs easier to discover, complete, verify, and repeat in Cactus 2.0?
3. Scale: can more users, organizations, and deals succeed with less manual onboarding and support?
4. Monetization: does the improved path increase evaluation-to-paid conversion, retention, expansion, and ACV?

Executive decision

Cactus should build one connected growth system, not ten independent feature projects:

1. Let a qualified prospect reach a defensible result before paying.

2. Make the result trustworthy enough to adopt.
3. Turn screening and market intelligence into repeat habits.
4. Preserve the customer's Excel model and controls to win larger accounts.
5. Add governance, collaboration, and decision outputs to expand accounts.
6. Enter development and commercial segments only through paid, narrowly scoped bets.

The highest unconstrained business-impact projects are Trust Center, Model Studio, Instant Deal Screen, Market Intelligence, and Guided Evaluation. The first build sequence is different: shared measurement, Guided Evaluation, Trust Center, and Instant Deal Screen must land first because they unlock the funnel through which the larger enterprise and expansion bets produce revenue.

How the portfolio was prioritized

The Product Manager Skills framework points to a weighted score with strategic overrides, rather than pure RICE:

- Cactus is in early product-market-fit/scaling mode.
- There is rich qualitative evidence but uneven behavioral and revenue data.
- Several initiatives are platform or enterprise bets that RICE would underweight.
- Dependencies and time-to-revenue matter as much as raw impact.

The business-growth score uses:

- 35% revenue potential
- 20% adoption and retention impact
- 15% strategic leverage
- 10% user-experience impact
- 10% evidence confidence
- 10% delivery leverage, using effort as a directional proxy

This score is a prioritization input, not an ARR forecast or ROI calculation. Cactus does not yet have billing, CRM stage, product usage, and closed-won/lost data joined well enough to calculate feature ROI.

Business impact rank	Initiative	Growth score	Effort	Primary revenue pathway	Investment posture
1	Trust Center 2.0	95	L	Conversion, retention, enterprise win rate	Build now
2	Model Studio	89	XL	Enterprise ACV, pilot conversion, expansion	Validate now; build supported subset next
3	Instant Deal Screen	88	M	New-logo conversion, usage frequency, deal volume	Build now
4	Market Intelligence and Memory	87	XL	Retention, expansion, differentiated paid value	Validate now; build next
5	Guided Evaluation and Sage Activation	85	M	Evaluation-to-paid conversion, lower onboarding cost	Build now; first execution priority
6	Enterprise Workflow	83	L	Larger ACV, multi-user deployment, expansion	Design council now; build next
7	Scenario Lab	80	L	Engagement, seat expansion, retention	Instrument now; build next
8	Development Underwriter	77	XL	New segment ARR	Paid beta later

Business impact rank	Initiative	Growth score	Effort	Primary revenue pathway	Investment posture
9	Commercial Lease Intelligence	72	XL	New asset-class ARR	Narrow paid beta after validation
10	Decision Packages	71	M	Workflow completion, collaboration, expansion	Concierge validation; productize later

Strategy context

Product promise

Turn a messy real-estate deal package into a defensible, reusable decision while preserving the customer's model, evidence, and judgment.

Business outcomes

1. Acquire more paying customers: move value demonstration before the payment decision.
2. Increase activation and retention: get users to an approved or exported first deal and repeat the workflow.
3. Increase ARPA and ACV: support proprietary models, multiple users, controls, integrations, and premium workflows.
4. Create expansion revenue: monetize Market Intelligence, Model Studio, enterprise controls, and higher-volume usage.
5. Reduce cost to serve: decrease manual onboarding, model setup, support, and rework per activated account.
6. Expand the addressable market carefully: enter development and multi-tenant commercial only when paid demand clears explicit gates.

The growth and revenue loop

Workflow

- Sample deal or controlled own-model workspace -> Sage-guided first defensible deal
- Sage-guided first defensible deal -> Trust Center approval
- Trust Center approval -> Instant screening and repeat underwriting
- Instant screening and repeat underwriting -> Market Intelligence and Scenario Lab
- Market Intelligence and Scenario Lab -> Model Studio and Enterprise Workflow
- Model Studio and Enterprise Workflow -> Decision Packages and collaboration
- Decision Packages and collaboration -> Paid conversion, expansion, and proprietary memory
- Paid conversion, expansion, and proprietary memory -> Instant screening and repeat underwriting

Portfolio programs

Program A - Self-serve growth engine

Projects: Guided Evaluation and Sage Activation, Instant Deal Screen, shared growth telemetry.

Business job: Increase qualified evaluation volume, evaluation-to-paid conversion, and repeat deal activity while reducing onboarding labor.

Revenue metrics: New ARR, evaluation-to-paid conversion, sales-cycle time, onboarding cost per activated account, CAC payback once acquisition cost is joined.

Program B - Defensible underwriting core

Projects: Trust Center, Model Studio, Scenario Lab, Decision Packages.

Business job: Make Cactus trustworthy enough to replace manual rework, fit established firm workflows, and complete a decision inside the product.

Revenue metrics: Gross revenue retention, enterprise pilot-to-paid conversion, expansion ARR, product-related loss reasons, support and rework cost.

Program C - Compounding data and habit

Projects: Market Intelligence and Proprietary Memory.

Business job: Create a recurring workflow and proprietary-data advantage that improves every future deal.

Revenue metrics: Weekly active underwriting teams, accounts reusing approved comps, expansion MRR, retention by module-adoption cohort.

Program D - Enterprise and segment expansion

Projects: Enterprise Workflow, Development Underwriter, Commercial Lease Intelligence.

Business job: Increase ACV and expand the addressable market without turning Cactus into a CRM, construction-management product, or ARGUS clone.

Revenue metrics: Enterprise ACV, multi-user production deployments, connected workspaces, segment-specific paid beta ARR, gross margin by workflow.

Build sequence

Horizon 0 - Prove the plan and repair measurement (0-30 days)

This horizon is mandatory. It prevents the team from shipping a free trial or major enterprise bet without knowing whether it creates revenue.

Shared product and revenue telemetry

- Establish stable `account_id`, `organization_id`, `deal_id`, and user identity across product events.

- Join product usage to CRM opportunity, stage, source, segment, closed-won/lost reason, plan, ARR, and expansion.
- Add evaluation, approval, export, share, scenario, and paid-conversion events.
- Separate production, staging, development, internal, and support-impersonation traffic.
- Establish baselines for:
 - evaluation started -> meaningful action -> approved/exported result -> paid;
 - onboarding and support hours per activated account;
 - logo churn, revenue churn, GRR, NRR, ARPA, ACV, and expansion MRR;
 - product COGS per active deal and per activated account.

Guided Evaluation discovery and design

- Replace the generic trial question with a qualified Start guided evaluation journey.
- Prototype three entry goals: learn Cactus, screen a deal, and test my model.
- Test a sample-deal path and a controlled own-model compatibility path.
- Redesign Sage as a page-aware, deal-aware coach with a persistent checklist outside chat.
- Map navigation, unit mix, income and expense, node terminology, missing inputs, errors, and human escalation.

Parallel validation

- Trust Center: test material-exception prioritization with 8-10 analysts.
- Instant Deal Screen: concierge-screen 100 packages for 8-12 design partners.
- Model Studio: spike 20 anonymized workbooks and publish a supported-subset report.
- Market Intelligence: validate evidence cards and private-memory behavior on 30 deals.
- Enterprise: identify which security, permissions, integration, or API blocker is attached to the largest qualified pipeline.

Exit gate

Proceed to Horizon 1 only when:

- the evaluation and revenue funnel can be measured end to end;
- at least two Guided Evaluation prototypes have been usability tested;
- Trust and Instant Screen have narrow, validated vertical slices;
- Model Studio has explicit supported and unsupported boundaries.

Horizon 1 - Build the conversion and trust engine (30-90 days)

1. Guided Evaluation and Sage Activation

Rename the existing Guided Activation - First Defensible Deal project to Guided Evaluation and Sage Activation.

Hypothesis: Letting a qualified prospect complete a sample or controlled own-model evaluation with contextual Sage guidance will increase evaluation-to-paid conversion and reduce onboarding labor because only 27.7% of test-filtered

Cactus 2.0 signups currently reach a meaningful action within 14 days.

First release:

- website entry and goal selection;
- disposable evaluation workspace;
- guided multifamily sample deal;
- controlled own-model compatibility check;
- persistent four-step progress path;
- contextual Sage explanations, navigation, validation, and human handoff;
- visible action confirmation and audit trail;
- approved/exported first-result event;
- value summary and paid conversion handoff.

Proposed decision gate:

- Cactus 2.0 signup/evaluation-to-meaningful-action improves from 27.7% to at least 40%;
- approved/exported first-result tracking is reliable;
- onboarding/support hours per activated account decline;
- trust and error guardrails do not worsen.

Do not ship: an unlimited anonymous trial, an unbounded model upload promise, or a product tour detached from a real task.

2. Trust Center 2.0

Hypothesis: Prioritizing material exceptions and preserving reviewer decisions will improve conversion, retention, and enterprise confidence because users hesitate to delegate consequential underwriting to results they cannot verify.

First release:

- T-12 and rent-roll exception queue;
- materiality and confidence ranking;
- source drawer and competing values;
- approve, replace, defer, and escalate;
- downstream model impact;
- reviewer audit trail and tie-out summary.

Proposed decision gate:

- material review time declines by at least 30% in beta;
- escaped material-error rate does not increase;
- at least 10 accounts complete the review-to-approved-input workflow.

3. Instant Deal Screen

Hypothesis: A zero-setup, explainable package-to-go/no-go workflow will increase new-logo conversion and repeat usage because Cactus 1.0 Quick Analysis recorded 100,593 pageviews from 215 external identified users, while the comparable named Cactus 2.0 route recorded only 18 pageviews from 3 users. This is a directional cross-version migration gap, not a like-for-like funnel.

First release:

- upload and forwarded-email intake;
- one multifamily buy-box template;
- saved criteria;
- explainable pass, fail, uncertain, and missing-data states;
- promote to full underwriting without re-entry;
- outcome and false-negative review events.

Proposed decision gate:

- package-to-decision time falls materially against the concierge baseline;
- users screen multiple deals per active account;
- false-negative and missing-data guardrails are reviewed before automation expands.

Horizon 2 - Deepen paid value and enterprise fit (3-6 months)

4. Market Intelligence and Memory

Hypothesis: Reviewable comp evidence and reusable private memory will increase retention and expansion because Market Intelligence is already the broadest verified Cactus 2.0 custom workflow: 1,415 submitted messages from 85 external users across 50 organizations.

First release:

- rent-comp evidence cards for one asset class;
- source, freshness, relevance, and confidence;
- accept/reject with reason;
- attach comp to a material assumption;
- save and reuse customer-owned comps;
- short source-backed market memo.

Gate: demonstrate faster approved comp selection and repeated reuse in design-partner accounts before adding more sources or asset classes.

5. Model Studio

Hypothesis: A supported, formula-aware Excel roundtrip will increase enterprise pilot-to-paid conversion and ACV because firms will not abandon proprietary models or accept hard-coded exports.

First release:

- one workbook and one asset-class template;
- named-range and cell mapping;
- supported-formula preservation;
- unsupported construct detection;
- test run with real deal values;
- versioned template registry;
- tie-out and change report.

Gate: at least 80% of selected pilot workbooks tie out within an agreed tolerance, with no silent unsupported failures. Expand coverage only from observed paid demand.

6. Enterprise Workflow foundation

Hypothesis: Reusable governance and integration capabilities will increase ACV and production deployment because enterprise champions cannot expand Cactus without security, permissions, API, and external-system fit.

First release:

- project roles and least-privilege defaults;
- audit explorer;
- SSO configuration diagnostics;
- one connector selected from qualified pipeline;
- asynchronous API jobs, status, and webhooks;
- multi-user activation and deployment measurement.

Gate: enterprise design partners move from champion-only use to measurable multi-user production deployment.

7. Scenario Lab

Hypothesis: Bounded natural-language scenarios with visible diffs will increase engagement and account expansion because customers already use Sage for what-if questions but current scenario behavior is not directly measured.

First release:

- five common assumption changes;
- proposed change and model-impact preview;
- material-change confirmation;
- immutable approved base case;
- two-scenario comparison;
- save, restore, share, and scenario lineage.

Gate: at least 70% of the validated common scenario requests complete without analyst support, while invalid or silently misapplied changes remain below the agreed quality threshold.

Horizon 3 - Complete the workflow and run paid segment bets (6-12 months)

8. Decision Packages

Hypothesis: Turning approved deal state into source-linked decision material will increase collaboration, retention, and expansion because teams currently rebuild conclusions manually.

First step: concierge-generate packages for 20 completed deals before committing a product team.

First product slice after validation:

- one IC template;
- executive summary and five red-flag types;
- source-linked conclusions;
- editable preview;
- live read-only link and PDF export;
- stale-package warning.

Gate: reduce approved-underwriting-to-shared-package time by at least 50% in the validated audience while keeping unsupported claims at zero.

9. Development Underwriter paid beta

Hypothesis: A narrowly scoped development model will create new segment ARR because developers need cost, draw, stabilization, and role-specific inputs that acquisition models cannot represent.

First release:

- one multifamily ground-up archetype;
- hard and soft cost stack;
- percentage-based fees;
- straight-line draw;
- simple lease-up and stabilization;
- role-scoped cost entry;
- formula-aware export and tie-out.

Gate: 5-8 paid or contractually committed design partners complete and approve models without rebuilding them offline.

Boundary: underwriting, decision, and handoff - not invoice, vendor, change-order, or construction project management.

Horizon 4 - Conditional commercial expansion (12+ months or earlier only with paid pull)

10. Commercial Lease Intelligence

Hypothesis: Amendment-aware lease abstraction and a reviewable rollover schedule can create new office, retail, or industrial ARR without requiring full ARGUS parity.

First release:

- one asset class and lease type;
- base lease and amendment chain;
- critical-term abstraction;
- rent-roll discrepancy review;
- basic rollover and recovery schedule;
- supported Excel or ARGUS handoff.

Gate: choose the segment from qualified pipeline, benchmark at least 500 leases against human abstraction, define the accuracy threshold before development, and require paid beta commitment.

Shared platform dependencies

Workflow

- Stable deal/account identity + revenue telemetry -> Enterprise deployment measurement
- Source metadata + reviewer state -> Market Intelligence
- Source metadata + reviewer state -> Decision Packages
- Model parameter schema + versioning -> Scenario Lab
- Model parameter schema + versioning -> Development Underwriter
- Model parameter schema + versioning -> Commercial Lease Intelligence
- Identity + audit + async jobs -> Enterprise deployment measurement
- Identity + audit + async jobs -> Decision Packages

Recommended team topology

This plan assumes two cross-functional product pods plus a shared platform/data lane:

Pod 1 - Growth and workflow

- Guided Evaluation and Sage Activation
- Instant Deal Screen
- later: Market Intelligence and Decision Packages

Pod 2 - Trust and enterprise fit

- Trust Center
- Model Studio spike and MVP
- later: Enterprise Workflow and Scenario Lab

Shared lane

- product and revenue telemetry;
- deal/account identity;
- source/reviewer state;
- model parameter schema and versioning;
- Sage action contract, audit, async jobs, and quality evaluation.

If Cactus has only one delivery pod, preserve the same order and reduce concurrency:

1. measurement and Guided Evaluation slice;
2. Trust Center slice;
3. Instant Deal Screen;
4. Market Intelligence;
5. Model Studio;
6. Enterprise Workflow;
7. Scenario Lab;
8. Decision Packages;
9. Development beta;
10. Commercial beta.

Portfolio measurement system

Acquisition and activation

- qualified website visitors starting an evaluation;
- Cactus 2.0 evaluation workspace created;
- Cactus 2.0 meaningful underwriting action within 14 days;
- approved/exported first result;
- evaluation-to-paid conversion;
- sales-cycle time and loss reason;
- onboarding/support hours per activated account.

Retention and expansion

- Cactus 2.0 week-1 and week-4 retained accounts by cohort;
- cross-version migration rate for proven Cactus 1.0 jobs, beginning with Quick Analysis;
- logo churn and revenue churn;
- gross revenue retention and net revenue retention;
- expansion MRR by adopted module;

- active deals and returning underwriting teams per account;
- ARPA, ACV, seats, and active users per account.

Trust and quality

- time from extraction complete to approved inputs;
- material exceptions resolved;
- escaped material errors;
- false-negative screening rate;
- model tie-out rate;
- unsupported or invalid Sage changes;
- correction and export-rework rate.

Economics and guardrails

- product COGS per active deal;
- AI/extraction cost per approved deal;
- contribution margin by plan and workflow;
- support and model-onboarding cost;
- gross margin effect of data licenses, connectors, and model-processing workloads.

Quarterly operating cadence

1. Review revenue and retention by cohort and module.
2. Review each initiative's leading product metric and revenue outcome.
3. Re-score initiatives with Product, Design, Engineering, Sales, Customer Success, and Finance.
4. Continue, narrow, pause, or stop initiatives based on their decision gate.
5. Publish what moved, what was learned, what was stopped, and why.

Scores must not become automatic commitments. A high score without a validated problem, supported technical path, or measurable revenue connection is a discovery project - not a full build.

What not to build now

- An unlimited anonymous free trial.
- A generic Sage chatbot without page context, model controls, confirmation, or audit.
- Arbitrary Excel or VBA support.
- A replacement CRM or transaction-management product.
- Construction project management, invoice approval, or vendor management.
- Full ARGUS parity or simultaneous support for every commercial asset class.

- New modules whose usage cannot be joined to paid conversion, retention, or expansion.

Linear execution

The existing file `cactus_linear_projects_and_tickets_2026-07-27.csv` already contains six tickets for each of the ten product projects. Before importing or creating projects in Linear:

1. **Rename** Guided Activation - First Defensible Deal **to** Guided Evaluation and Sage Activation.
2. Add a shared Growth and Revenue Telemetry project.
3. Add Guided Evaluation tickets for website entry, sandbox/workspace creation, own-model compatibility, contextual Sage, human escalation, and paid conversion.
4. Add project properties for revenue pathway, business-impact rank, execution horizon, effort, confidence, and decision gate.
5. Keep projects in discovery until their entry gate is satisfied; do not move all 60 tickets into active delivery.

Required executive inputs before financial ROI claims

To turn this plan into a contribution-margin and ROI model, Cactus still needs:

- current MRR and ARR;
- ARPA and ACV by segment and plan;
- new, expansion, churned, and contraction MRR;
- logo churn, revenue churn, GRR, and NRR;
- gross margin and product COGS;
- sales pipeline and closed-won/lost data tied to product blockers;
- onboarding, custom-model, and support labor cost;
- qualified pipeline attached to enterprise, development, and commercial requests.

Until those inputs are joined, project revenue effects should be labeled hypotheses, not forecasts.

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Cactus PostHog behavioral evidence

Analysis date: 2026-07-27

Executive readout

- Cactus 1.0 external identified monthly activity declined from 118 users in July 2025 to 24 in partial July 2026.
- Cactus 2.0 external identified monthly activity grew from 33 users in December 2025 to 64 in June 2026; partial July recorded 52.
- Cactus 2.0: activation is the largest measured current-product growth constraint: 91 of 329 test-filtered signups reached chat, extraction, or template creation within 14 days (27.7%).
- Cactus 2.0: mature identified-user cohorts retained at 43.3% in week 1 and 18.6% in week 4; cohort sizes are small.
- Cactus 2.0: Market Analysis is the broadest verified current custom workflow: 1,415 messages from 85 external users across 50 organizations.
- Cross-version: Quick Analysis was the strongest measured Cactus 1.0 habit, with 100,593 pageviews from 215 external identified users, while only 3 users reached the named Cactus 2.0 quick-analysis route.
- Cactus 2.0: Trust and Excel pipelines show meaningful use plus reliability work: extraction converted 85.3% of test-filtered users within one day; Excel extraction recorded 327 successes and 15 failures.

How product versions are used

- Cactus 1.0 is the legacy behavior benchmark: it identifies proven user jobs and historical usage intensity.
- Cactus 2.0 is the current-product baseline: it measures whether the refactor improves discoverability, completion, trust, repeat use, and scalability.
- Cross-version comparisons identify migration gaps. They are directional unless the underlying events, routes, populations, and time windows have been verified as equivalent.
- A strong Cactus 1.0 signal can justify preserving a job, but it cannot establish that the Cactus 2.0 implementation is successful.
- A weak Cactus 2.0 signal does not automatically invalidate the job when Cactus 1.0 demand was strong; it can instead indicate missing parity, discoverability, or instrumentation.
- Declining Cactus 1.0 activity during migration is not, by itself, evidence of declining demand or churn.
- Active-user counts across Cactus 1.0 and 2.0 must not be added until cross-project identity resolution prevents double counting.

Opportunity evidence

Rank	Opportunity	Behavioral evidence	Strength
1	Trust Center 2.0 - Exception-First Document Review	Cactus 2.0: 29 of 34 test-filtered users completed extraction within one day (85.3%); median conversion was 40.8 seconds and average was 105.5 seconds. Across raw pipeline events, 514 extraction completions covered 41 users. Confidence verification averaged 0.766 confidence and 948.8 source matches across 1,362.1 verified facts per run. Workflow routes logged 1,422 exceptions affecting 35 external users, although a benign ResizeObserver issue dominates the grouped error list.	Strong, narrow cohort
2	Market Intelligence - Evidence-Backed Comps and Memory	Cactus 2.0 external usage: market-analysis generated 1,415 submitted messages from 85 users across 50 organizations (16.6 messages per user), making it the broadest verified custom workflow in the current product.	Strong and broad
3	Instant Deal Screen - Intake-to-Go/No-Go	Cactus 1.0 external identified usage: Quick Analysis generated 100,593 pageviews from 215 users over the last twelve months (467.9 views per user). The comparable Cactus 2.0 quick-analysis route recorded only 18 pageviews from 3 users, indicating a valuable legacy job that has not visibly migrated.	Strong legacy habit; migration gap
4	Model Studio - Formula-Aware Excel Roundtrip	Cactus 2.0 recorded 342 Excel extraction completions across 31 users: 327 successes and 15 failures (95.6% event-level success), with successful runs averaging 147.6 seconds. Several workbook-related LLM steps show 18-40% failure rates, and active error tracking includes worksheet-name collisions.	Strong technical signal; moderate reach
5	Guided Activation - First Defensible Deal	Only 91 of 329 test-filtered Cactus 2.0 signups reached a meaningful action (chat submission, extraction start, or template creation) within 14 days: 27.7% conversion. Converters reached value quickly (median 8.0 minutes), so the primary problem is getting more new users onto the successful path. Mature identified-user cohorts retained at 43.3% in week 1 and 18.6% in week 4, with small denominators.	Strong growth constraint
6	Scenario Lab - Conversational Sensitivity With Controls	Cactus 2.0 chat is established: 90 of 106 test-filtered submitters received an assistant completion within ten minutes (84.9%), with a 2.1-second median and 16.2-second average. No dedicated scenario-created, confirmed-change, or scenario-compared event exists, so scenario demand is inferred from chat and transcripts rather than directly measured.	Strong adjacent behavior; missing scenario event
7	Enterprise Workflow - Integrations, Roles, and Governance	Cactus 2.0 logged 36 organization invitations from 18 organizations and 23 acceptances across 15 organizations. The roughly 64% event-level acceptance ratio is directional because sent and accepted events are not joined by invite.	Early but measurable
8	Development Underwriter - Cost, Draw, and Stabilization	Cactus 2.0 LLM steps tied to development-style tables show elevated failure: Unit Mix Table 40.9%, Project Stages Table 26.3%, and several cost-table steps 22-26%. Samples are small and do not establish a large active development cohort.	Technical pain; limited adoption evidence
9	Decision Packages - Source-Linked IC, Lender, and Client Outputs	Cactus 2.0 recorded 39 chat-share events across 13 users, while external report routes reached only 2 users. Package generation, approval, live views, export, and downstream decision events are not instrumented.	Weak or under-instrumented
10	Commercial Lease Intelligence - Multi-Tenant Underwriting	The test-filtered Cactus 2.0 rent-roll funnel included 4 triggering users and 2 completions within 30 minutes (50%; 74.9-second median among converters). Raw completion events covered only 6 users, too small for a broad build decision.	Very early sample

Measurement caveats

- The PostHog projects use UTC and person-on-events mode; person properties reflect event-time values.

- Known internal users were excluded from SQL aggregates using the company email domain when available. PostHog's configured test-account filter was used for funnel and retention queries.
- Cactus 1.0 pageviews contain substantial anonymous and automated traffic. Legacy adoption statements therefore use identified users and exclude known internal users.
- Several Cactus 2.0 server-side events label `environment` as `dev` even when paired with production-facing activity. Environment is not used as a production filter until instrumentation is corrected.
- Event taxonomy discovery omitted some older or dynamic events/properties that bounded SQL queries found. Every metric names its actual event population.
- The data-warehouse schema endpoint returned `INVALID_ARGUMENT` in both projects. Analyses used verified event/property schemas plus bounded HogQL and typed funnel/retention queries.
- `semantic_matching_prompt_too_long` has no discoverable properties, so its 790 occurrences across 22 raw users cannot be reliably attributed to a workflow.
- Scenario creation, deal approval/export, decision-package generation, dead clicks, and several revenue outcomes are not consistently instrumented. Absence of evidence is not evidence of low demand.
- July 2026 is partial through July 27. Small cohorts, especially rent-roll and development workflows, should not drive broad investment without a design-partner gate.

PostHog projects

- Cactus 1.0: internal PostHog project; aggregate results are reported above.
- Cactus 2.0: internal PostHog project; aggregate results are reported above.

Machine-readable metrics are in `cactus_posthog_metrics_2026-07-27.csv`.

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Cactus Product & Design Opportunity Portfolio

Research date: 2026-07-27

Executive recommendation

Cactus should concentrate the product strategy on one promise: turn a messy deal package into a defensible decision while preserving the customer's model, evidence, and judgment. The transcript corpus does not support a generic "more AI" roadmap. It supports a connected workflow with four layers:

1. Trust the inputs: prioritize and resolve material exceptions.
2. Reach a decision faster: screen, model, and test scenarios without re-entry.
3. Fit the firm: preserve Excel, templates, roles, and integrations.
4. Compound the work: reuse approved evidence, assumptions, and outputs on the next deal.

The integrated ranking places Trust Center 2.0 first, elevates Market Intelligence from seventh to second because it is already the broadest verified Cactus 2.0 workflow, and keeps Instant Deal Screen third because Quick Analysis was the strongest legacy usage habit. For delivery, start Trust Center, Instant Deal Screen, and Guided Activation now; begin Market Intelligence evidence-card validation and the Model Studio technical spike in parallel.

Research basis

- 946 unique Sybill transcripts from externally labeled meetings.
- 113,688 normalized speaker turns and 5.72 million total words.
- 56,829 non-Cactus speaker turns and 2.21 million external-speaker words after excluding known Cactus team names.
- 25 meetings classified as likely internal/advisor conversations were excluded from weighted recommendation evidence.
- Evidence was mined at meeting level so a talkative call did not count as dozens of independent votes.
- Thematic counts are directional qualitative evidence, not survey percentages; one meeting can support several opportunities.
- PostHog Cactus 1.0 and Cactus 2.0 projects were analyzed for verified events, external identified usage, activation, workflow adoption, funnels, retention, reliability, and instrumentation gaps.
- Cactus 1.0 external identified monthly activity declined from 118 users in July 2025 to 24 in partial July 2026; Cactus 2.0 grew from 33 in December 2025 to 64 in June 2026, with 52 in partial July.

The analysis followed the linked Product Manager Skills methods: voice-of-customer themes were framed as needs rather than feature requests; an opportunity-solution structure was used before selecting solutions; a weighted scoring model combined customer voice with verified behavior; delivery was split into vertical, testable slices.

Important temporal limitation

The manifest does not contain meeting dates. Some calls clearly describe earlier product states, while Cactus's current public pages already describe source trails, document reconciliation, custom templates, screening, scenario analysis, Market Intelligence, Proprietary Memory, Excel export, API access, SSO/SAML, and new-development support. PostHog reduces this uncertainty by showing which workflows are currently used, but incomplete or inconsistent instrumentation still prevents precise revenue attribution and end-to-end deal completion measurement.

Prioritization method

Integrated impact score uses 20% user experience + 20% adoption + 20% revenue + 10% strategic leverage + 15% transcript evidence + 15% PostHog behavioral evidence. Each dimension is scored 1-5, with small or missing cohorts explicitly penalized. Effort remains separate because the purpose is to identify impact before engineering sizing.

Rank	Project	Impact	Effort	Roadmap	Explicit-signal meetings	PostHog evidence
1	Trust Center 2.0 - Exception-First Document Review	97	L	Now	276	Strong, narrow cohort
2	Market Intelligence - Evidence-Backed Comps and Memory	96	XL	Next	266	Strong and broad
3	Instant Deal Screen - Intake-to-Go/No-Go	94	M	Now	99	Strong legacy habit; migration gap
4	Model Studio - Formula-Aware Excel Roundtrip	93	XL	Next	184	Strong technical signal; moderate reach
5	Guided Activation - First Defensible Deal	92	M	Now	453	Strong growth constraint
6	Scenario Lab - Conversational Sensitivity With Controls	84	L	Next	216	Strong adjacent behavior; missing scenario event
7	Enterprise Workflow - Integrations, Roles, and Governance	79	L	Next	213	Early but measurable
8	Development Underwriter - Cost, Draw, and Stabilization	74	XL	Later	149	Technical pain; limited adoption evidence
9	Decision Packages - Source-Linked IC, Lender, and Client Outputs	68	M	Later	79	Weak or under-instrumented
10	Commercial Lease Intelligence - Multi-Tenant Underwriting	67	XL	Later	31	Very early sample

Reading the evidence: "Explicit-signal meetings" contain a rule-based opportunity match plus request, pain, adoption, revenue, or trust language. Counts are not additive. PostHog strength reflects current behavioral breadth, funnel quality, and instrumentation confidence - not just raw event volume.

What PostHog changed

- Market Intelligence moved from #7 to #2: Cactus 2.0 market-analysis chat recorded 1,415 messages from 85 external users across 50 organizations.
- Instant Deal Screen stayed in the top three: Cactus 1.0 Quick Analysis logged 100,593 pageviews from 215 external identified users, but the named Cactus 2.0 route reached only 3 users - an urgent migration gap.
- Guided Activation remains Now: only 91 of 329 test-filtered Cactus 2.0 signups reached a meaningful underwriting action within 14 days (27.7%).
- Model Studio remains a major bet but requires a reliability gate: 327 of 342 Excel extraction completion events succeeded, with 147.6-second average latency and concentrated workbook-step failures.

- Commercial Lease Intelligence moved to #10: the test-filtered rent-roll funnel contained only 4 triggering users and 2 completions.

Full behavioral evidence and caveats: `cactus_posthog_product_evidence_2026-07-27.md` and `cactus_posthog_metrics_2026-07-27.csv`.

Strategic problem framing

Look inward: Cactus's risk is defining success as feature coverage - more asset classes, models, data, and AI behaviors - while users evaluate whether they can reach and defend a decision in their own workflow.

Look outward: The problem is most acute for small teams processing high deal volume, analysts inheriting complex models, principals who need quick what-if answers, and enterprise teams that cannot move data outside controlled systems. The status quo benefits manual services and spreadsheet expertise, but it limits product adoption and organizational scale.

Reframed problem: CRE teams struggle to delegate underwriting because messy inputs, model variation, evidence review, and stakeholder handoff remain disconnected. Even a fast AI result creates rework when the team cannot verify it, fit it into its model, or reuse the approved decision.

How might we help a CRE team move from package receipt to a defensible, reusable decision with minimal re-entry while keeping professionals in control?

Opportunity-solution tree

Outcome A: Reduce package-to-defensible-decision time

- └ Trust material inputs
 - └ Trust Center 2.0
- └ Reject low-fit deals earlier
 - └ Instant Deal Screen
- └ Answer what-if questions safely
 - └ Scenario Lab

Outcome B: Increase activation and repeat usage

- └ Reach first value without a live call
 - └ Guided Activation
- └ Preserve the firm's model
 - └ Model Studio
- └ Complete the decision workflow
 - └ Decision Packages

Outcome C: Win larger accounts and expand ARPA

- └ Fit enterprise systems and controls
 - └ Enterprise Workflow
- └ Compound proprietary market knowledge
 - └ Market Intelligence & Memory

Outcome D: Expand the addressable workflow

- └ Ground-up development
 - └ Development Underwriter
- └ Multi-tenant office/retail/industrial
 - └ Commercial Lease Intelligence

Project briefs

1. Trust Center 2.0 - Exception-First Document Review

Posture: Deepen an existing core capability

Target user: Acquisitions analyst, principal reviewer, lender/credit reviewer

Impact / effort / roadmap: 97/100 · L · Now

Directional evidence: 353 matching meetings; 276 with explicit signals; 59 post-sale or working-session meetings.

PostHog baseline (Strong, narrow cohort): Cactus 2.0: 29 of 34 test-filtered users completed extraction within one day (85.3%); median conversion was 40.8 seconds and average was 105.5 seconds. Across raw pipeline events, 514 extraction completions covered 41 users. Confidence verification averaged 0.766 confidence and 948.8 source matches across 1,362.1 verified facts per run. Workflow routes logged 1,422 exceptions affecting 35 external users, although a benign ResizeObserver issue dominates the grouped error list.

Problem: Analysts will not delegate consequential underwriting work to AI when the remaining errors are difficult to find, conflicts are not prioritized, or review decisions disappear after export.

How might we: How might we surface and resolve the few facts that can materially change the deal before those facts flow into the model?

Proposed experience: Add an exception-first Review Queue to the existing source-linked extraction flow. Rank facts by financial materiality and confidence; show the source snippet, competing values, model impact, and reviewer state in one fact card. Let analysts approve, replace, defer, or escalate a fact, then preserve that decision in an audit log and tie-out report.

Design direction: Primary flow: deal intake summary -> material exceptions -> fact drawer -> resolve -> publish approved inputs. Design for calm review: risk hierarchy, side-by-side sources, keyboard navigation, persistent filters, empty/loading/failed states, and a visible distinction between extracted facts, assumptions, and human approvals.

MVP slice: T-12 + rent-roll conflicts for multifamily; materiality rules; source drawer; approve/replace action; reviewer audit trail; tie-out summary.

Out of scope: Autonomous approval, a universal diligence checklist, and every document type in the first release.

Primary metric: Median minutes from extraction complete to approved model inputs.

Secondary and guardrail metrics: High-risk exceptions resolved; correction rate; reviewer agreement; support tickets about wrong values; percentage of deals exported without rework.

Validation: Prototype with 8-10 analysts using real conflicted packages; beta with 10 accounts and compare review time/error escape rate with the current workflow.

Dependencies: Extraction confidence metadata, source storage, model impact graph, reviewer identity, event analytics.

Key risk: A noisy queue will reduce trust. Start with a narrow, calibrated definition of material exceptions and expose why each item was prioritized.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

2. Market Intelligence - Evidence-Backed Comps and Memory

Posture: Deepen current Market Intelligence and Proprietary Memory

Target user: Market analyst, acquisitions professional, principal

Impact / effort / roadmap: 96/100 · XL · Next

Directional evidence: 433 matching meetings; 266 with explicit signals; 69 post-sale or working-session meetings.

PostHog baseline (Strong and broad): Cactus 2.0 external usage: market-analysis generated 1,415 submitted messages from 85 users across 50 organizations (16.6 messages per user), making it the broadest verified custom workflow in the current product.

Problem: Finding comps is time-consuming, but unverified or stale comps are worse. Teams need to understand why a comp supports an assumption and preserve trusted private knowledge for the next deal.

How might we: How might we turn market data into reviewable evidence that improves every future underwriting rather than another data feed?

Proposed experience: Create comp evidence cards with source, freshness, relevance, and confidence. Let users accept/reject a comp, attach it to an assumption, add a private comp, and save the decision into firm memory. Generate a short source-backed market memo.

Design direction: Primary flow: suggested comps/map -> inspect evidence -> accept/reject -> attach to assumption -> save to memory. Separate vendor data, public records, customer-owned data, and inference visually. Support map and table workflows equally.

MVP slice: Rent-comp evidence cards for one asset class; freshness/relevance indicators; accept/reject; attach to market rent; save customer-owned comps.

Out of scope: Owning every market-data source or replacing specialist GIS/data products.

Primary metric: Percentage of material market assumptions with approved supporting evidence.

Secondary and guardrail metrics: Time to comp set; comp acceptance rate; private comps saved/reused; repeat-deal time savings; Market Intelligence weekly actives.

Validation: Compare current comp selection with evidence cards on 30 deals; measure agreement, time, and the reasons users reject recommendations.

Dependencies: Data licenses, entity resolution, freshness metadata, map performance, firm memory schema.

Key risk: More data can create false confidence. Make source limitations explicit and optimize for defensibility, not volume of data points.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

3. Instant Deal Screen - Intake-to-Go/No-Go

Posture: Complete and simplify an existing screening capability

Target user: Acquisitions lead, solo sponsor, broker, mortgage originator

Impact / effort / roadmap: 94/100 · M · Now

Directional evidence: 182 matching meetings; 99 with explicit signals; 44 post-sale or working-session meetings.

PostHog baseline (Strong legacy habit; migration gap): Cactus 1.0 external identified usage: Quick Analysis generated 100,593 pageviews from 215 users over the last twelve months (467.9 views per user). The comparable Cactus 2.0 quick-analysis route recorded only 18 pageviews from 3 users, indicating a valuable legacy job that has not visibly migrated.

Problem: Teams spend too long deciding whether a deal deserves full underwriting and cannot process the volume required to find the few opportunities worth pursuing.

How might we: How might we turn an incoming package into an explainable go/no-go decision with almost no setup?

Proposed experience: Create a zero-touch intake queue fed by upload, forwarded email, and API. Apply saved buy-box criteria, show pass/fail reasons with source evidence, and let the user promote a promising screen into a full underwriting without re-entry.

Design direction: Primary flow: receive deal -> processing queue -> screen card -> inspect reasons -> reject/archive or promote. Optimize for scanning many deals: table/card density, batch actions, processing status, explicit missing-data states, and saved filters.

MVP slice: Forwarded email or upload intake; saved criteria; one multifamily screening template; explainable pass/fail; promote to full underwriting.

Out of scope: A full CRM, broker relationship management, and transaction execution.

Primary metric: Median time from package receipt to recorded go/no-go decision.

Secondary and guardrail metrics: Deals screened per active account; screens promoted; offers submitted; return frequency; processing completion rate.

Validation: Concierge-screen 100 real packages for 8-12 design partners, compare decision time and agreement with their current process, then release a narrow queue MVP.

Dependencies: Document intake, criteria schema, Quick Analysis, job status, deduplication, email security.

Key risk: A false negative can hide a good deal. Show uncertainty and missing data; avoid binary automation when the evidence is incomplete.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

4. Model Studio - Formula-Aware Excel Roundtrip

Posture: Turn current custom-model support into a productized moat

Target user: Senior underwriter, acquisitions team lead, enterprise model owner

Impact / effort / roadmap: 93/100 · XL · Next

Directional evidence: 289 matching meetings; 184 with explicit signals; 40 post-sale or working-session meetings.

PostHog baseline (Strong technical signal; moderate reach): Cactus 2.0 recorded 342 Excel extraction completions across 31 users: 327 successes and 15 failures (95.6% event-level success), with successful runs averaging 147.6 seconds. Several workbook-related LLM steps show 18-40% failure rates, and active error tracking includes worksheet-name collisions.

Problem: Firms have proprietary Excel models, formulas, and review conventions. A static export or one-off services mapping creates adoption friction and makes teams re-enter or re-check work.

How might we: How might we let a firm keep its trusted model while Cactus supplies mapped, source-backed inputs and a reliable roundtrip?

Proposed experience: Productize model onboarding with a mapping wizard, supported-formula report, template registry, version history, and formula-aware Excel export. Generate a tie-out report showing what mapped, what did not, and which values changed.

Design direction: Primary flow: upload model -> inspect workbook -> map input/output cells -> test with one deal -> resolve unmapped items -> publish template. Use a two-pane mapping interface, confidence/status chips, preview values, and reversible mapping edits. Make unsupported formulas/macros explicit before launch.

MVP slice: One workbook, one asset-class template, named-range/cell mapping, supported formula preservation, formula-aware export, and tie-out report.

Out of scope: Arbitrary VBA/macro execution, every Excel feature, and replacing Excel as a general-purpose modeling environment.

Primary metric: Percentage of pilot models that tie out within agreed tolerance.

Secondary and guardrail metrics: Time to onboard a model; repeat use of mapped templates; export rework rate; enterprise pilot-to-paid conversion.

Validation: Run a technical spike on 20 anonymized customer workbooks, then a concierge pilot with five firms before committing to self-serve mapping.

Dependencies: Workbook parser, calculation engine boundaries, template schema, versioning, security review.

Key risk: The long tail of Excel behavior can consume the roadmap. Publish a supported subset, detect unsupported constructs early, and preserve a service-assisted path.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

5. Guided Activation - First Defensible Deal

Posture: Redesign the first-run experience around time-to-value

Target user: Trial user, new analyst, account champion onboarding teammates

Impact / effort / roadmap: 92/100 · M · Now

Directional evidence: 509 matching meetings; 453 with explicit signals; 115 post-sale or working-session meetings.

PostHog baseline (Strong growth constraint): Only 91 of 329 test-filtered Cactus 2.0 signups reached a meaningful action (chat submission, extraction start, or template creation) within 14 days: 27.7% conversion. Converters reached

value quickly (median 8.0 minutes), so the primary problem is getting more new users onto the successful path. Mature identified-user cohorts retained at 43.3% in week 1 and 18.6% in week 4, with small denominators.

Problem: New users encounter unfamiliar underwriting concepts, required inputs, and processing states before they experience a trustworthy result. Dedicated training helps, but it does not scale or rescue self-serve trial drop-off.

How might we: How might we help a new user complete one defensible deal without needing a live walkthrough?

Proposed experience: Ask for role and goal, then offer a guided sample deal or a real-deal path. Use a short contextual checklist, progressive disclosure, clear processing status, recoverable errors, and a first-value summary that explains what Cactus did.

Design direction: Primary flow: choose goal -> sample or real package -> guided review -> adjust one assumption -> export/share. Avoid tours detached from work. Guidance should be dismissible, resume across sessions, and adapt to analyst versus principal roles.

MVP slice: Role/goal selection; one guided multifamily sample; four-step checklist; upload progress and recovery; first-deal completion event.

Out of scope: Gamification, a large academy rebuild, and deep persona personalization.

Primary metric: Percentage of trials reaching an approved/exported first deal.

Secondary and guardrail metrics: Time to first value; checklist completion; trial-to-paid conversion; onboarding support hours; day-7 return rate.

Validation: Instrument the current funnel, usability-test two first-run prototypes, then A/B test guided sample versus the current start state.

Dependencies: Identity/role model, sample data, job status, event analytics, help content.

Key risk: Guidance can feel like extra work to experts. Provide a direct expert path and let users switch paths without losing progress.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

6. Scenario Lab - Conversational Sensitivity With Controls

Posture: Unify current sensitivity and AI capabilities

Target user: Principal, analyst, broker, lender reviewing an underwriting

Impact / effort / roadmap: 84/100 · L · Next

Directional evidence: 344 matching meetings; 216 with explicit signals; 72 post-sale or working-session meetings.

PostHog baseline (Strong adjacent behavior; missing scenario event): Cactus 2.0 chat is established: 90 of 106 test-filtered submitters received an assistant completion within ten minutes (84.9%), with a 2.1-second median and 16.2-second average. No dedicated scenario-created, confirmed-change, or scenario-compared event exists, so scenario demand is inferred from chat and transcripts rather than directly measured.

Problem: Decision-makers want to ask natural what-if questions, but unconstrained chat can silently change the wrong input, violate model rules, or lose the path back to the approved base case.

How might we: How might we make scenario creation as easy as asking a question while keeping every change explicit, reversible, and model-valid?

Proposed experience: Combine structured sensitivity controls with a conversational command bar. Parse the requested change, preview affected assumptions and outputs, require confirmation for material changes, and save a versioned scenario with a side-by-side diff.

Design direction: Primary flow: ask/change -> preview diff -> confirm -> compare -> save/share. Pair chat with visible controls and constraint feedback. The base case is immutable; scenario lineage and changed inputs remain visible throughout review.

MVP slice: Natural-language changes for five common assumptions; validation preview; two-scenario comparison; save/restore; audit entry.

Out of scope: A general-purpose chatbot and autonomous investment recommendations.

Primary metric: Scenario creation completion rate without analyst support.

Secondary and guardrail metrics: Scenarios per active deal; invalid-change rate; time to answer what-if questions; shared scenario views; weekly active principals.

Validation: Wizard-of-Oz 30 common what-if requests, establish a safe command grammar, then prototype with principals before enabling free-form commands.

Dependencies: Model parameter schema, calculation engine, validation rules, scenario versioning.

Key risk: Chat implies more freedom than the model safely supports. Preview every change and prefer bounded commands until accuracy is demonstrated.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

7. Enterprise Workflow - Integrations, Roles, and Governance

Posture: Productize the Max-plan platform layer

Target user: Enterprise admin, underwriting manager, IT/security buyer, external reviewer

Impact / effort / roadmap: 79/100 · L · Next

Directional evidence: 329 matching meetings; 213 with explicit signals; 53 post-sale or working-session meetings.

PostHog baseline (Early but measurable): Cactus 2.0 logged 36 organization invitations from 18 organizations and 23 acceptances across 15 organizations. The roughly 64% event-level acceptance ratio is directional because sent and accepted events are not joined by invite.

Problem: Larger teams need Cactus inside their existing data and approval environment. One-off integrations, coarse permissions, or unclear audit/security controls slow procurement and limit deployment beyond a champion.

How might we: How might we let enterprise teams connect, govern, and observe Cactus without turning Cactus into another system of record?

Proposed experience: Deliver project-level RBAC, admin-visible audit logs, self-service SSO configuration, API jobs/webhooks, and prioritized external storage connectors beginning with SharePoint or Google Drive. Keep external systems authoritative.

Design direction: Primary surfaces: admin security center, role matrix, integration setup, sync/job monitor, audit explorer. Use explicit permission previews, least-privilege defaults, test connection states, and actionable failure messages.

MVP slice: Project roles; audit log; one external storage connector; asynchronous API job status and webhook; SSO configuration diagnostics.

Out of scope: Rebuilding CRM, document management, or full transaction management.

Primary metric: Enterprise pilots reaching multi-user production deployment.

Secondary and guardrail metrics: Time through security/onboarding; connected workspaces; API jobs; active users per account; enterprise win rate and expansion.

Validation: Run a design council with five enterprise prospects/customers; identify the connector and permission set that blocks the largest qualified pipeline.

Dependencies: Identity/tenant model, audit events, API platform, connector framework, security review.

Key risk: Integrations can become bespoke services. Select connectors by pipeline and reusable architecture; publish support levels and ownership boundaries.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

8. Development Underwriter - Cost, Draw, and Stabilization

Posture: Deepen a current launch workflow without becoming project management

Target user: Developer, development analyst, construction lead, capital partner

Impact / effort / roadmap: 74/100 · XL · Later

Directional evidence: 242 matching meetings; 149 with explicit signals; 55 post-sale or working-session meetings.

PostHog baseline (Technical pain; limited adoption evidence): Cactus 2.0 LLM steps tied to development-style tables show elevated failure: Unit Mix Table 40.9%, Project Stages Table 26.3%, and several cost-table steps 22-26%. Samples are small and do not establish a large active development cohort.

Problem: Ground-up underwriting requires percentage-based fees, cost timing, draw schedules, lease-up/stabilization, and role-specific inputs that acquisition models do not handle.

How might we: How might we let a mixed finance/construction team build and defend a development underwriting without exposing every input or recreating execution software?

Proposed experience: Create configurable development archetypes with a cost stack, percentage formulas, schedule/curve builder, debt draw logic, and role-scoped input views. Export a source-backed model and variance/tie-out report.

Design direction: Primary flow: choose archetype -> build cost stack -> set schedule -> define financing -> model lease-up -> review returns. Use linked visual timelines, formula chips, dependency warnings, and separate construction-input from returns-review views.

MVP slice: One multifamily ground-up archetype; hard/soft cost stack; percentage fees; straight-line draw; simple lease-up curve; role-scoped cost entry.

Out of scope: Invoice approvals, vendor management, change orders, and full construction execution.

Primary metric: Development models completed and approved without offline rebuild.

Secondary and guardrail metrics: Time to model; template reuse; cost-entry participation; export tie-out; development pilot conversion and expansion ARR.

Validation: Run a paid/design-partner beta with 5-8 developers and validate one archetype before adding curves, debt structures, and asset classes.

Dependencies: Schedule engine, formula relationships, permissions, debt model, Excel export.

Key risk: Scope can drift into Northspyre-style development management. Keep the boundary at underwriting, decision, and handoff.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

9. Decision Packages - Source-Linked IC, Lender, and Client Outputs

Posture: A fast-follow workflow-completion win

Target user: Analyst preparing a decision, principal, broker, lender, client

Impact / effort / roadmap: 68/100 · M · Later

Directional evidence: 136 matching meetings; 79 with explicit signals; 28 post-sale or working-session meetings.

PostHog baseline (Weak or under-instrumented): Cactus 2.0 recorded 39 chat-share events across 13 users, while external report routes reached only 2 users. Package generation, approval, live views, export, and downstream decision events are not instrumented.

Problem: After underwriting, teams still re-read the model, identify red flags, and manually reformat the conclusion for investment committee, lenders, or multiple clients.

How might we: How might we turn approved underwriting into a concise, source-backed decision package without creating another static artifact that immediately goes stale?

Proposed experience: Generate a configurable executive summary, key assumptions, red flags, scenario table, and source links from approved deal state. Support firm/client templates, a live review link, and export bundles.

Design direction: Primary flow: choose audience/template -> preview generated sections -> edit/approve -> share live or export. Preserve the link from every conclusion to its model input/source and display when the package is stale relative to the deal.

MVP slice: One IC template; executive summary; five red-flag types; source links; editable preview; PDF export and live read-only link.

Out of scope: A general presentation editor and every client format in the first release.

Primary metric: Median time from approved underwriting to shared decision package.

Secondary and guardrail metrics: Packages generated; sections edited; live views; stale-package warnings; repeated use; support for custom formats.

Validation: Generate concierge packages for 20 completed deals, observe edits and audience needs, then productize the common 80%.

Dependencies: Approved deal state, risk flags, scenario data, document generation, permissions.

Key risk: Generated narrative can overstate certainty. Limit claims to approved inputs and label inference, assumptions, and missing evidence.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

10. Commercial Lease Intelligence - Multi-Tenant Underwriting

Posture: Advance from lease due diligence to a narrow commercial underwriting beta

Target user: Office/retail/industrial acquisitions analyst and broker

Impact / effort / roadmap: 67/100 · XL · Later

Directional evidence: 56 matching meetings; 31 with explicit signals; 7 post-sale or working-session meetings.

PostHog baseline (Very early sample): The test-filtered Cactus 2.0 rent-roll funnel included 4 triggering users and 2 completions within 30 minutes (50%; 74.9-second median among converters). Raw completion events covered only 6 users, too small for a broad build decision.

Problem: Multi-tenant commercial deals require lease/amendment abstraction, recovery structures, rollover assumptions, and portfolio-scale review. This is a painful segment, but depth expectations are set by ARGUS.

How might we: How might we turn many leases and amendments into a reviewable rollover model without asking the analyst to start from scratch?

Proposed experience: Batch-ingest leases, build an amendment-aware abstraction, reconcile it with the rent roll, and generate a tenant rollover/recovery schedule. Export the approved schedule into supported Excel/ARGUS handoffs.

Design direction: Primary flow: upload lease set -> processing monitor -> tenant abstraction queue -> rent-roll comparison -> rollover schedule -> export. The UI must handle partial success, duplicate/amended documents, tenant identity, and hundreds of exceptions.

MVP slice: One asset class and lease type; batch upload; base lease + amendment chain; critical term abstraction; rent-roll discrepancy report; basic rollover schedule.

Out of scope: All commercial asset classes, full ARGUS parity, and lease administration.

Primary metric: Analyst hours to produce an approved tenant rollover schedule.

Secondary and guardrail metrics: Critical-term accuracy; exception rate; leases per completed job; export rework; paid beta conversion.

Validation: Choose office or industrial based on qualified pipeline; benchmark 500 leases against human abstractions before promising full underwriting.

Dependencies: Document hierarchy, tenant/entity resolution, recovery rules, schedule engine, export integration.

Key risk: Commercial lease variability can overwhelm the team. Enter through a narrow segment with measurable accuracy and a human-review workflow.

Public-edition evidence basis: Aggregate transcript counts are reported above. Direct customer quotes, participant names, timestamps, and private Sybill recording links are omitted.

Now / Next / Later plan

Now - prove the core promise

- Trust Center 2.0: narrow MVP for material T-12/rent-roll exceptions.
- Instant Deal Screen: explainable package-to-go/no-go workflow.
- Guided Activation: first-defensible-deal path and funnel instrumentation.
- Model Studio: technical spike only; do not promise arbitrary workbook support yet.
- Shared analytics foundation: deal identity, organization identity, activation, approval/export, scenario, and reliability events.

Next - deepen workflow fit

- Market Intelligence evidence cards and private-memory validation, using the existing 50-organization behavior as the design-partner pool.
- Model Studio MVP for one asset class and supported Excel subset.
- Scenario Lab with bounded natural-language changes and versioning.
- Enterprise Workflow foundation: project roles, audit log, one connector, API job status.

Later - make explicit segment bets

- Development Underwriter beta for one archetype.
- Decision Packages only after generation, edit, share, export, and view instrumentation establishes demand.
- Commercial Lease Intelligence beta for one asset/lease slice.

This sequence is intentionally Now/Next/Later, not a delivery commitment. Engineering must size the enabling architecture and current production gaps before dates are assigned.

Portfolio dependencies

```
Source metadata + reviewer state
├─ Trust Center
├─ Decision Packages
└─ Market Intelligence evidence

Model parameter schema + versioning
├─ Model Studio
├─ Scenario Lab
├─ Development Underwriter
└─ Commercial Lease Intelligence

Identity + audit events + async jobs
├─ Guided Activation analytics
├─ Enterprise Workflow
└─ External sharing
```

Competitive implications

- Cactus's current public positioning already emphasizes source-backed facts, reviewer control, Excel-ready output, Market Intelligence, and Proprietary Memory. The highest-impact roadmap should make those claims visibly true in end-to-end workflows, not add disconnected feature labels.
- redIQ publicly offers a proprietary valuation model and QuickSync to place standardized data into customer Excel templates. Formula-aware, lower-friction model interoperability is therefore a competitive requirement, not an edge case.
- ARGUS Enterprise remains the depth benchmark for lease-by-lease modeling, market-leasing assumptions, scenario analysis, configurable reports, and Excel connectivity. Cactus should enter commercial through a narrow automation/review wedge rather than promise immediate parity.
- Dealpath owns pipeline, CRM, integrations, reporting, and adjacent AI. Cactus should integrate with systems of record rather than spend core capacity rebuilding a CRM.
- Northspyre spans development acquisition through construction, draw management, and stabilization. Cactus's development boundary should remain underwriting, decision support, and handoff.

What not to build now

- A generic chat assistant without model controls, lineage, and confirmation.
- A full CRM or transaction-management replacement.
- Full construction project management, invoice approval, or vendor workflows.
- Simultaneous deep support for every commercial asset class.
- A hard break from Excel; customer evidence and competitors both argue for interoperability.

Measurement plan

Create a shared product scorecard before development begins:

- North star: median time from package receipt to approved/shareable decision.
- Activation: percentage of new accounts completing one approved or exported real/sample deal.
- Trust: material corrections after approval and percentage of key inputs with reviewed evidence.
- Habit: deals screened/underwritten per active account and weekly returning underwriting teams.
- Revenue: pilot-to-paid conversion, enterprise win rate, expansion ARR, and loss reasons tied to missing workflow capabilities.
- Efficiency guardrail: onboarding/support hours per activated account.
- Quality guardrail: false-negative screening rate and escaped material extraction/model errors.

Decision gates before committing engineering capacity

1. Confirm current production gaps for each project; several capabilities already exist publicly.
2. Join transcript evidence to CRM stage, account value, closed/lost reason, and current customer status.
3. Repair the analytics gaps: consistent environment labels, stable deal/account identity, scenario events, approval/export/share outcomes, dead-click coverage, and revenue joins.
4. Size each project with Engineering and identify shared platform work.
5. Select 5-10 design partners across prospect, new customer, active customer, and enterprise segments.
6. Re-score the portfolio collaboratively; scores are decision inputs, not automatic commitments.

Artifacts

Machine-readable evidence indexes, account-level research, and Linear import files remain internal and are not included in this public edition.

Sources and framework references

- [Product Manager Skills](#)
- [Cactus CRE underwriting product](#)
- [Cactus pricing and current plan capabilities](#)
- [Cactus AI-powered due diligence](#)
- [redIQ product and QuickSync positioning](#)
- [ARGUS Enterprise](#)
- [Dealpath CRM and platform positioning](#)
- [Northspyre commercial development platform](#)
- [Linear importer field documentation](#)
- Cactus 1.0: internal PostHog project; aggregate results are reported above.
- Cactus 2.0: internal PostHog project; aggregate results are reported above.

Research caveats

- Meeting date, account status, company, ARR, opportunity stage, closed/lost outcome, and asset class were not consistently present in the manifest.
- External labeling includes some product partners, vendors, advisors, and working sessions; title-based classification reduced but cannot eliminate this noise.
- Transcript language reflects both independent customer voice and reactions to a demo. Seller narration was excluded by known speaker name, but leading-question effects remain.
- Evidence counts are theme-detection outputs and should not be presented as statistically representative market incidence.
- PostHog cohorts are small, partially migrated, and unevenly instrumented. Known internal users were excluded when event-time email was available; typed funnels used the configured test-account filter.
- Several server events label environment as `dev`, taxonomy discovery omitted older dynamic events, and the data-warehouse schema endpoint failed. Behavioral findings therefore distinguish strong direct signals from weak or missing instrumentation.
- Effort estimates are preliminary t-shirt sizes inferred from scope, not engineering commitments.

Public redacted edition. This document contains aggregate product research. Direct customer identities, verbatim quotes, private recordings, account-level data, and authenticated analytics links are intentionally excluded.

Cactus Growth and Revenue Roadmap

Planning date: 2026-07-27

Evidence: Cactus 1.0 legacy behavior, Cactus 2.0 current-product behavior, aggregate Sybill research, and business-growth prioritization.

This is an outcome-based roadmap. Scores and horizons are decision inputs, not delivery promises or financial forecasts.

Execution sequence

0. Growth and Revenue Telemetry

- Business-impact rank: Enabler
- Program: Shared platform
- Growth score: Enabler
- Estimated effort: M
- Investment posture: Build immediately
- Horizon: Horizon 0
- Revenue mechanism: Revenue attribution; lower decision risk
- Primary metric: Evaluation-to-paid funnel coverage
- PostHog evidence scope: Cross-version: create a stable bridge from Cactus 1.0 legacy behavior through Cactus 2.0 usage to CRM and billing outcomes
- Decision gate: Stable account, organization, deal, approval, export, paid-conversion, CRM, billing, and environment fields are queryable end to end
- Dependencies: Identity; CRM and billing joins; PostHog taxonomy; data ownership

1. Guided Evaluation and Sage Activation

- Business-impact rank: 5
- Program: Self-serve growth engine
- Growth score: 85
- Estimated effort: M
- Investment posture: Build now
- Horizon: Horizons 0-1
- Revenue mechanism: Evaluation-to-paid conversion; lower onboarding cost
- Primary metric: Percentage of evaluations reaching an approved or exported first result
- PostHog evidence scope: Cactus 2.0 current-product baseline: 91 of 329 test-filtered signups reached a meaningful action within 14 days (27.7%)

- Decision gate: Cactus 2.0 meaningful-action conversion improves from 27.7% to at least 40%; support hours decline; quality guardrails do not worsen
- Dependencies: Growth telemetry; sample data; Sage context and action contract; job status; payment handoff

2. Trust Center 2.0 - Exception-First Document Review

- Business-impact rank: 1
- Program: Defensible underwriting core
- Growth score: 95
- Estimated effort: L
- Investment posture: Build now
- Horizon: Horizons 0-1
- Revenue mechanism: Conversion; retention; enterprise win rate
- Primary metric: Median minutes from extraction complete to approved model inputs
- PostHog evidence scope: Cactus 2.0 current-product extraction and confidence-verification evidence
- Decision gate: Material review time declines at least 30%; escaped material errors do not increase; 10 beta accounts complete review to approved inputs
- Dependencies: Source metadata; confidence; model impact graph; reviewer state; audit

3. Instant Deal Screen - Intake-to-Go/No-Go

- Business-impact rank: 3
- Program: Self-serve growth engine
- Growth score: 88
- Estimated effort: M
- Investment posture: Build now
- Horizon: Horizons 0-1
- Revenue mechanism: New-logo conversion; repeat usage; deal throughput
- Primary metric: Median time from package receipt to recorded go or no-go decision
- PostHog evidence scope: Cross-version: strong Cactus 1.0 Quick Analysis habit with a directional Cactus 2.0 migration gap
- Decision gate: Package-to-decision time improves against concierge baseline; multiple screens per active account; false-negative review is operational
- Dependencies: Document intake; buy-box criteria; deduplication; job state; promotion to underwriting

4. Market Intelligence - Evidence-Backed Comps and Memory

- Business-impact rank: 4
- Program: Compounding data and habit
- Growth score: 87
- Estimated effort: XL

- Investment posture: Validate now; build next
- Horizon: Horizons 0 and 2
- Revenue mechanism: Retention; expansion; differentiated paid value
- Primary metric: Percentage of material market assumptions with approved supporting evidence
- PostHog evidence scope: Cactus 2.0 current-product Market Analysis adoption
- Decision gate: Design partners select approved comps faster and reuse private comps before source or asset-class expansion
- Dependencies: Data licenses; entity resolution; freshness; source metadata; firm-memory schema

5. Model Studio - Formula-Aware Excel Roundtrip

- Business-impact rank: 2
- Program: Defensible underwriting core
- Growth score: 89
- Estimated effort: XL
- Investment posture: Technical spike now; supported subset next
- Horizon: Horizons 0 and 2
- Revenue mechanism: Enterprise ACV; pilot conversion; expansion
- Primary metric: Percentage of pilot models tying out within agreed tolerance
- PostHog evidence scope: Cactus 2.0 current-product Excel extraction usage and reliability
- Decision gate: At least 80% of selected pilot workbooks tie out; zero silent unsupported failures
- Dependencies: Workbook parser; model schema; formula boundaries; template registry; security

6. Enterprise Workflow - Integrations Roles and Governance

- Business-impact rank: 6
- Program: Enterprise and segment expansion
- Growth score: 83
- Estimated effort: L
- Investment posture: Design council now; build next
- Horizon: Horizon 2
- Revenue mechanism: Higher ACV; multi-user deployment; expansion
- Primary metric: Enterprise pilots reaching multi-user production deployment
- PostHog evidence scope: Cactus 2.0 current-product invitation and collaboration evidence
- Decision gate: Qualified enterprise design partners progress from champion-only use to measurable multi-user production
- Dependencies: Tenant identity; roles; audit; SSO; connector framework; API jobs

7. Scenario Lab - Conversational Sensitivity With Controls

- Business-impact rank: 7
- Program: Defensible underwriting core
- Growth score: 80
- Estimated effort: L
- Investment posture: Instrument now; build next
- Horizon: Horizon 2
- Revenue mechanism: Engagement; retention; seat expansion
- Primary metric: Scenario completion without analyst support
- PostHog evidence scope: Cactus 2.0 current-product chat behavior; dedicated scenario events are missing
- Decision gate: At least 70% of validated common scenario requests complete without analyst help and within agreed quality limits
- Dependencies: Model parameter schema; calculation engine; validation; versioning; Sage action contract

8. Decision Packages - Source-Linked IC Lender and Client Outputs

- Business-impact rank: 10
- Program: Defensible underwriting core
- Growth score: 71
- Estimated effort: M
- Investment posture: Concierge validation; productize later
- Horizon: Horizon 3
- Revenue mechanism: Workflow completion; collaboration; expansion
- Primary metric: Median time from approved underwriting to shared decision package
- PostHog evidence scope: Cactus 2.0 current-product share evidence; package workflow is under-instrumented
- Decision gate: Twenty concierge packages establish the common 80%; product slice cuts creation time at least 50% with zero unsupported claims
- Dependencies: Approved deal state; risk flags; scenario data; source links; permissions; document generation

9. Development Underwriter - Cost Draw and Stabilization

- Business-impact rank: 8
- Program: Enterprise and segment expansion
- Growth score: 77
- Estimated effort: XL
- Investment posture: Paid beta later
- Horizon: Horizon 3
- Revenue mechanism: New segment ARR
- Primary metric: Development models completed and approved without offline rebuild
- PostHog evidence scope: Cactus 2.0 current-product technical-failure evidence; active cohort remains unproven

- Decision gate: Five to eight paid or committed design partners complete and approve the narrow archetype without offline rebuild
- Dependencies: Schedule engine; model relationships; permissions; debt model; Excel export

10. Commercial Lease Intelligence - Multi-Tenant Underwriting

- Business-impact rank: 9
- Program: Enterprise and segment expansion
- Growth score: 72
- Estimated effort: XL
- Investment posture: Narrow paid beta after validation
- Horizon: Horizon 4
- Revenue mechanism: New asset-class ARR
- Primary metric: Analyst hours to produce an approved tenant rollover schedule
- PostHog evidence scope: Cactus 2.0 current-product rent-roll evidence; sample is very small
- Decision gate: Segment selected from qualified pipeline; at least 500 leases benchmarked; accuracy threshold and paid commitments established before build
- Dependencies: Document hierarchy; tenant resolution; recovery rules; schedule engine; export integration

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